Weakly Supervised Semantic Segmentation With Consistency-Constrained Multiclass Attention for Remote Sensing Scenes

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***Abstract*— Obtaining image-level class labels for remote sensing (RS) images is a relatively straightforward process, sparking significant interest in weakly supervised semantic segmentation (WSSS). However, RS images present challenges beyond those encountered in generic WSSS, including complex backgrounds, densely distributed small objects, and considerable scale variations. To address the above issues, we introduce a consistency-constrained multiclass attention model, noted as CocoaNet. Specifically, CocoaNet endeavors to capture both semantic correlation and class distinctiveness using a global–local adaptive attention mechanism, which integrates the self-attention to model global correlation, complemented by a local perception branch that intensifies focus on local regions. The resulting class-specific attention weights and the patch-level pairwise affinity weights are employed to optimize the initial class activation maps (CAMs). This mechanism proves highly effective in mitigating interclass interference and managing the distribution of densely clustered small objects. Moreover, we invoke a consistency constraint to rectify activation inaccuracy. By utilizing a Siamese structure for the mutual supervision of features extracted from images at different scales, we address substantial scale variations in RS scenes. Simultaneously, a class contrast loss is adopted to enhance the discriminativeness of class-specific features. Departing from the conventional CAM optimization, which is rather complex and time-consuming, we harness the prior knowledge from the generic segment anything model (SAM) to design a joint optimization strategy (JOS) that refines target boundaries and further promotes discriminative visual features. We validate the effectiveness of our proposed approach on three benchmark datasets in multiclass RS scenarios, and the experimental results demonstrate that our model yields promising advancements compared to state-of-the-art methods.**

***Index Terms*— Consistency constraint, global–local adaptive attention, weakly supervised semantic segmentation (WSSS).**

Manuscript received 12 March 2024; accepted 20 April 2024. Date of publication 23 April 2024; date of current version 6 May 2024. This work was supported by the National Natural Science Foundation of China under Grant 62202283 and Grant 62372284. *(Corresponding author: Dan Zeng.)*

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Digital Object Identifier 10.1109/TGRS.2024.3392737

I. Introduction

With the rapid advancement of high spatial resolution satellite imaging, there has been an exponential increase in the volume of high-resolution remote sensing (RS) images. These images play a pivotal role in a wide range of RS applications, including cloud detection [1], [2], land resource analysis [3], urban target localization [4], [5], and environmental protection [6]. Semantic segmentation, which involves pixel-wise classification, greatly enhances the understanding of these images facilitating relevant applications. The adoption of deep learning methods has led to significant advancements in RS semantic segmentation. However, fully supervised (FS) semantic segmentation often demands a substantial number of pixel-level annotations, particularly for high-resolution RS images. This process can be time-consuming and necessitates a wealth of expert knowledge and field research.

Weakly supervised semantic segmentation (WSSS) has emerged as a cost-effective approach, primarily relying on accessible forms of supervision, including image-level class labels, points, scribbles, and bounding boxes. In this work, we focus on investigating the utility of image-level class labels. The majority of existing methods primarily employ these labels to generate class activation maps (CAMs), which serve to provide initial target localization and generate pseudo masks. These masks subsequently act as pixel-wise ground truth for training an FS counterpart. For example, Wang et al. [8] propose a consistency regularization approach for CAM predictions. They subject images to various transformations and apply self-supervision to ensure correct activations. Ahn and Kwak [9], to further refine CAMs, employ a local response mechanism that propagates to nearby regions belonging to the same semantic entity. This approach predicts semantic affinities between adjacent image coordinates. In addition, Gao et al. [10] segment images into a series of patch tokens for spatial embedding, fully leveraging the self-attention mechanism in the visual transformer to emphasize the complete object extent of CAMs.

While WSSS has made significant progress for natural images, existing methods exhibit limitations when applied to RS images, especially in scenarios with diverse semantic concepts. As shown in Fig. 1, in comparison to natural images, RS images possess unique characteristics: 1) high spatial resolution, offering exceptional clarity with ground

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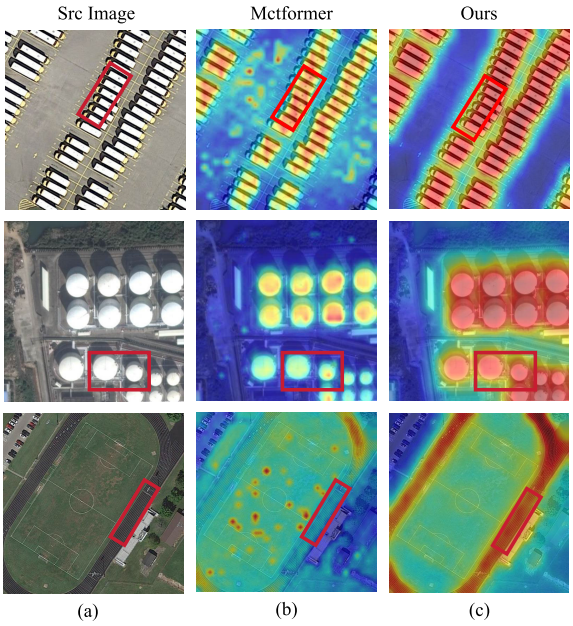


Fig. 1. (a) Source image. (b) Baseline method (Mctformer [7]). (c) Our approach CocoaNet. The first row illustrates densely distributed small targets. The second row demonstrates significant variability in object scales. The third row showcases the category confusion and susceptibility to misactivation. Our proposed model proves effective in addressing these issues.

details, while also containing densely distributed small-sized objects; 2) the object scales exhibit significant variability; and 3) Category confusion and easily prone to misactivation, making it difficult to distinguish target categories. Efforts have been made to incorporate these features into segmentation models. Zhou et al. [11] propose a prototype-based model to capture semantic similarities and differences between images. To address class confusion and background interference, they also introduce the single-label classification as an auxiliary task to facilitate training. In [12], a self-supervised Siamese network is introduced, featuring explicit pixel-level constraints. Leveraging a dynamic multiscale fusion module, this approach enhances feature expressiveness and positioning accuracy, particularly for small and densely distributed objects. Recognizing that CAMs typically activate only the most discriminative regions, Wang et al. [13] propose a scale-invariant multilevel context aggregation network to expand CAMs with finer boundaries and more comprehensive scopes.

In the context of complex multiclass scenes in RS imagery, research on WSSS remains relatively limited. Existing works tend to inherit frameworks from WSSS models designed for natural images. These models often lack a focus on emphasizing the discriminative aspects between different classes, which leads to confusion in the generated CAMs. Furthermore, the majority of them rely on conventional convolutional structures. While these structures effectively activate the most discriminative parts, interference from backgrounds often prevents the corresponding entire regions from being adequately delineated.

To better address challenges posed by RS images under limited supervision, our approach seeks to underscore the subtle correlations between intricate visual details and multiple semantic concepts that manifest within the given scene. We develop a multiclass token-based framework for generating pseudo masks with consistency-constrained attention, noted as CocoaNet. Fig. 2 presents the overall framework. During Training Phase 1, a two-stream structure is employed to train the CAM localization network from two image perspectives. The obtained class-specific attention weights and the patch-level pairwise affinity weights are used to optimize the initial CAMs. In the inference stage, a multiscale strategy is applied to generate the final CAMs, followed by iterative threshold filtering to obtain the initial pseudo masks. The segment anything model (SAM) branch is combined to generate semantic-free segments, and the joint optimization strategy (JOS) is utilized to optimize the final pseudo mask. During Training Phase 2, by referring to [12], pseudo masks serve as the supervision during the second-stage training of the DeepLabV1 with ResNet38 as the backbone. We dynamically establish suitable attention mechanisms from both global and local perspectives, further complemented by a scale consistency constraint and prior knowledge derived from a generic foundation model. Specifically, to fully explore the semantic dependencies among different classes while preserving their distinct characteristics, we employ a global–local adaptive attention mechanism to extract class-specific CAMs through the interactions between class and patch tokens. In addition to the conventional self-attention branch, which primarily captures long-range correlations, we introduce a parallel local perception branch (LPB) designed to enhance the focus on local regions by leveraging convolutional operations. We derive class-specific attention weights and patch-level pairwise affinity weights to optimize the initial CAMs. This mechanism mitigates the interclass interference and effectively addresses the distribution of densely clustered small objects.

Moreover, to compensate for the limited supervision from image-level labels and address scale variations, we introduce a consistency constraint by utilizing a Siamese structure to mutually supervise CAMs obtained at different scales. This constraint encourages pixel-level convergence to rectify respective activation errors. Simultaneously, we incorporate a class contrast loss that encourages each class token to exhibit self-similarity while being dissimilar to others, thereby enhancing distinctiveness among different categories during training. Finally, departing from the complex and time-consuming conventional CAM postprocessing, we harness the prior knowledge from the generic foundation model (SAM). We establish a JOS that capitalizes on the segments generated by SAM in the absence of accompanying class labels. By incorporating domain expertise specific to RS, we enhance the precision of pseudo mask boundaries and fine-tune their semantic distributions based on these class-agnostic segments. We validate the effectiveness of our proposed approach on three benchmark datasets in multiclass RS scenarios, our model surpasses state-of-the-art (SOTA) methods, achieving the overall best performance. In summary, the main contributions are as follows.

1) We design a global–local adaptive attention mechanism that facilitates the exploration of semantic dependencies between different classes while maintaining their distinctive characteristics. Combining self-attention with an LPB, it effectively mitigates interclass interference and

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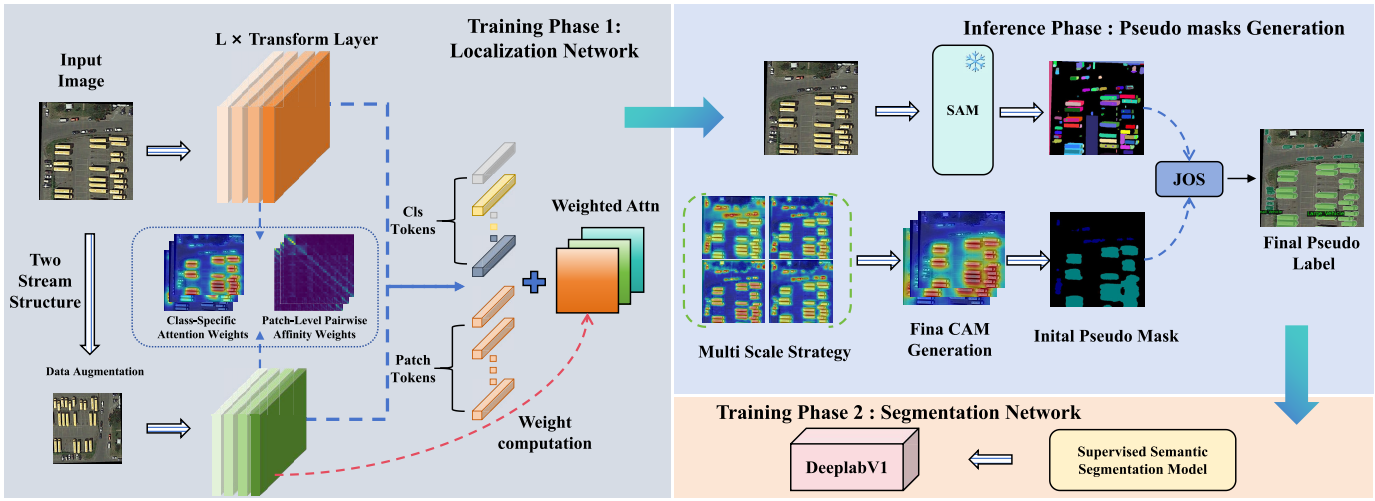


Fig. 2. CocoaNet overall architecture, our proposed framework for obtaining high-quality pseudo masks based on image-level labels. The framework integrates a dual-stream structure during Training Phase 1, capturing semantic correlations and class distinctiveness using a global–local adaptive attention mechanism. The subsequent stages involve multiscale strategies for generating final CAMs, iterative threshold filtering, and leveraging SAM and JOS branches for refining the pseudo masks. Training Phase 2 involves utilizing pseudo-labels to train the DeepLabV1 segmentation network with ResNet38 as the backbone, contributing to the overall effectiveness of the proposed approach.

enhances the segmentation of densely clustered small objects.

2) We introduce a consistency constraint, enabling mutual supervision between images at different scales through Siamese structure, addressing RS scale variations. The class contrast loss is adopted to further enhance the distinctiveness among different categories.
3) We propose to utilize prior knowledge from the generic foundation model to craft a CAM optimization strategy that refines target boundaries and adjust their semantic distributions by referring to class-agnostic segments.

II. Related Works

A. Semantic Segmentation in RS Scenes

Semantic segmentation tasks based on aerial and satellite images are crucial in various RS applications. Early works using sliding windows and candidate regions are time-consuming, requiring a substantial number of redundant calculations [14], [15], [16], [17]. Deep neural network-based models have emerged due to the availability of large-scale training data, including the U-Net architecture and its variants [18], [19], [20], [21]. Diakogiannis et al. [22] integrate multiple tasks within the ResU-Net framework. Each residual building block incorporates multiple parallel dilated convolutions to capture contextual information at different scales, which exhibits advanced performance in super-high-resolution segmentation for a single temporal phase. Priyanka et al. [23] extract multilevel features by integrating the perception module, modified residual blocks, and a dense global spatial pyramid pooling with the U-Net framework. This allows for the parallel extraction of both local and global perception. In conventional U-Net networks, a significant amount of semantic information is lost due to the pooling operation. MFNet [24] introduces a cross-type feature fusion that utilizes encoded information to supplement the lost details in the decoding process, enhancing the exchange and fusion of information across different levels of feature maps.

Traditional networks commonly use backbones such as FCN and U-Net, but they may fall short in fully exploring contextual information. Transformer-based methods excel in capturing long-range correlations [25], [26], [27]. For example, DC-Swin [28] integrates the Swin Transformer with a densely connected feature aggregation decoder to extract contextual information and enhance resolution for generating semantic masks. Meng et al. [29] further design a class-guided block based on the Swin Transformer to construct the decoder, using stacked convolutional layers for the extraction of category-based semantic features from the input image. These features are then jointly processed with those extracted from the Swin Transformer encoder. FS methods are limited by their need for extensive pixel-level annotations, which are time-consuming and labor-intensive. As a result, weakly supervised methods strive for cost-effective segmentation, reducing the dependence on detailed pixel-level annotations and offering promising prospects for RS semantic segmentation.

B. Image-Level WSSS

Depending on the available types of supervision, which range from simple to complex, WSSS methods can be categorized as follows: image-level class labels [30], [31], [32], [33], points [34], [35], [36], scribbles [37], [38], [39], and bounding boxes [40], [41]. We concentrate on image-level WSSS due to the ease and robustness of obtaining class labels using off-the-shelf classification models. Depending on the approach, WSSS for natural images is classified into two main types: single stage and multistage. Single-stage methods employ unified models with end-to-end training. Araslanov and Roth [42] improve CAMs by integrating score map with semantic mask using normalized global weighted pooling, with the goal of enhancing semantic integrity. Subsequently, they apply pixel-adaptive convolution to derive segmentation masks. SLRNet [43] utilizes cross-view self-supervised learning to predict complementary low-rank representations

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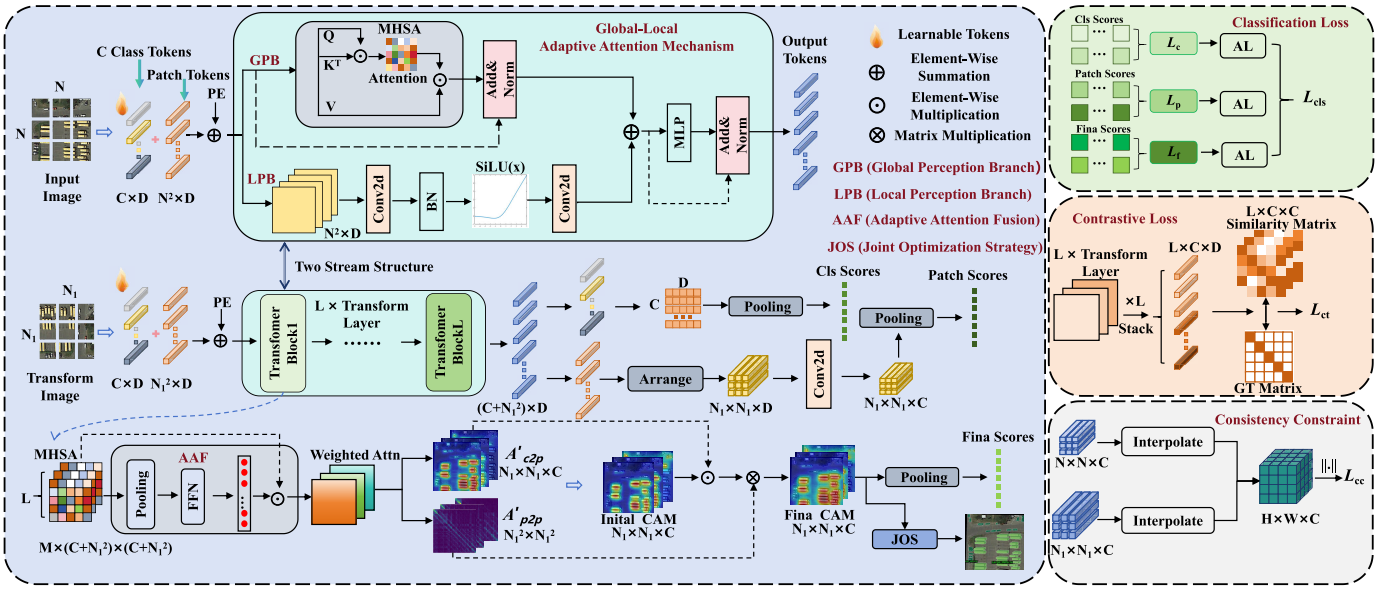


Fig. 3. Detailed implementation of the overall framework of CocoaNet. It begins by converting the input image into a series of patch tokens. Utilizing C additional class tokens, where C is the number of classes, these class tokens are concatenated with patch tokens, augmented with position embeddings (PEs), and then processed through consecutive L transformer encoding layers. Each layer consists of a global–local adaptive attention mechanism, integrating a GPB modeling global correlations, complemented by an LPB emphasizing local regions. In addition, a two-stream structure is introduced to supervise the generated CAMs. Initial CAMs are generated by applying a convolution layer to patch tokens. Subsequently, adaptive attention is employed to generate dynamic weights from all self-attention maps. Finally, the initial CAMs are optimized with class-specific attention weights and patch-level pairwise affinity weights. The loss function comprises three components: classification loss, contrastive loss, and consistency constraint.

from different views. This involves jointly decomposing learned representations from multiple views into submatrices, mitigating the impact of noise in the generated low-rank representation while simultaneously capturing stable semantics across different views.

Simultaneously handling the dual-branch tasks of localization and pixel-wise classification in a single-stage network often results in ineffective feature synchronization across branches. Maintaining a balance between pseudo masks generated in intermediate stages and the final segmentation results becomes crucial. Multistage methods adopt a step-by-step approach to tackle these challenges. These methods first generate high-quality CAMs and subsequently leverage these pseudo masks to train the final segmentation networks. EPS [44] integrates weak supervision from image-level labels and saliency maps, considerably improving target boundary accuracy via joint optimizing strategies. Typically, conventional CAMs tend to focus on small target regions. Lee et al. [45] eliminate the final activation function to alleviate information bottlenecks. They also introduce global nondiscriminative region pooling, filtering out values from spatial locations with CAM scores below a threshold to improve information transmission from nondiscriminative regions to the classification network. In CAM, each pixel might respond to multiple categories within the same receptive field. Chen et al. [46] utilize Softmax cross-entropy (CE) loss to refine the converged CAM, distinguishing pixel responses into distinct categories, thereby reducing ambiguity in the pseudo masks.

Image-level WSSS is less prevalent in intricate RS scenarios. In contrast to natural images, RS images possess unique attributes: intricate backgrounds, high spatial resolution, densely distributed small objects, and significant variability in object scales. Zhou et al. [11] propose a prototype-based model aimed at capturing semantic similarities and differences among RS images. They include single-label classification as an auxiliary task during training to mitigate class confusion and background interference. In [12], a self-supervised Siamese network is proposed, generating equivalent activation areas in different sizes of the same RS image. They eliminate the global average pooling and utilize the SoftMax to obtain a pixel confidence map, emphasizing explicit pixel-level constraints. Recognizing that CAMs typically activate only the most discriminative regions, Wang et al. [13] propose a scale-invariant multilevel context aggregation network to enhance CAMs with finer boundaries and broader scopes. Auxiliary classification branches are adopted to deepen the semantic understanding of low-level features and shorten the gradient backpropagation path. We also opt in the two-stage pipeline and establish a multiclass token-based framework for generating pseudo masks.

C. Postprocessing in WSSS

In multistage WSSS, postprocessing is commonly used to refine CAMs for obtaining high-quality pseudo masks. Ahn and Kwak [9] employ a local response mechanism to improve CAMs by propagating information to neighboring regions that share the same semantic entity, predicting semantic connections between adjacent image coordinates. Ahn et al. [47] identify confident seed regions of the target class from attention maps generated by a classification model, subsequently propagating these seeds to delineate entire instance regions with accurate boundaries. Although postprocessing operations enhance the quality of pseudo masks, they may potentially slow down the entire training process and require frequent hyperparameter adjustments. With the emergence of generic

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foundation models, the SAM [48] has gained significant attention. SAM can generate high-quality masks even without explicit semantic guidance. In the field of natural images, several studies propose combining SAM with WSSS [49], [50], [51], employing various types of weak labels to provide cues for SAM. For RS images, we argue that appropriately leveraging the class-agnostic generalization capability of generic foundation models allows for facilitating the efficient optimization of pseudo masks.

III. Methodology

In this article, we present a novel WSSS approach customized for RS images. While existing research primarily focuses on urban and rural scenes with fewer categories, our approach surpasses these limitations and demonstrates applicability in complex RS scenarios involving intricate backgrounds, densely distributed small objects, and significant scale variations. Section III-A offers an in-depth introduction to the overall framework. Section III-B presents the details of global–local adaptive attention mechanism, intending to capture semantic dependencies among various categories while maintaining their distinct features simultaneously. In Section III-C, we introduce a consistency constraint employing a Siamese structure to supervise CAMs acquired at different scales. Section III-D utilizes prior knowledge from the generic foundation model (SAM) to formulate a JOS. Finally, Section III-E introduces the integrated loss function.

A. Base Framework

Fig. 3 illustrates the detailed implementation of the proposed framework aimed at emphasizing the subtle correlations between complex visual details and various semantic concepts within the scene. Our design draws the inspiration from the Mctformer [7] as the base framework. The MCTformer captures class-specific attention by learning interactions among multiple class tokens, enabling the generation of CAMs with improved class discriminability. We also construct the framework utilizing multiclass tokens, which dynamically establishes adequate attention mechanisms from global and local perspectives, supported by a scale consistency constraint and leveraging prior knowledge from a generic foundation model.

To begin with, for each stream, we segment the input image into nonoverlapping patches at a specific scale, mapping them into patch tokens. Subsequently, we combine these patch tokens with class tokens across multiple categories to create input tokens, directing them through a series of transformer encoding layers. Each layer comprises a multihead self-attention (MHSA) mechanism and a dual-branch structure dedicated to local perception. This sequence enables the derivation of class-specific attention weights and patch-level pairwise affinity weights, culminating in the final optimized CAMs. During the training phase, we enforce a consistency constraint by mutually supervising CAMs obtained at various scales using a two-stream Siamese structure.

Moreover, we apply diverse loss functions to both scales of output tokens and multistage CAMs. In contrast to traditional methods that rely only on MLP for basic classification, our loss functions integrate refined multistage classification loss, consistency constraint, and class contrast loss. Further details are elaborated in Section III-B–III-E.

B. Global–Local Adaptive Attention Mechanism

The input image $I \in \mathbb{R}^{H\times W\times 3}$ undergoes partitioning into nonoverlapping patches within a single stream of the Siamese network, where H and W represent the height and width dimensions of the input image, respectively. s indicates the step size for dividing patches, while $D = 3 \times s^2$ denotes the patch embedding dimension resulting from flattening patches into 1-D sequences. In conventional Transformer models, patch tokens are traditionally combined with the class token, integrating positional information. However, this necessitates individual class tokens to simultaneously encompass semantic information from various categories, posing a potential risk of interference between them. As a solution, multiple class tokens are formulated, each capturing features unique to its corresponding category. The set of input tokens, $\boldsymbol{T} \in \mathbb{R}^{(C+N^2)\times D}$, where C represents the total number of classes and N stands for the number of tokens along each spatial dimension, is subsequently forwarded to the global–local adaptive attention mechanism, which incorporates a dual-branch local perception structure along with a global perception branch (GPB).

1) Local Perception Branch: In RS scenes, characterized by intricate backgrounds and densely scattered small objects, enhancing the perception and comprehension of local information within the image is imperative. This improvement is vital for refining the modeling capabilities concerning particular regions or contexts. Considering that class tokens inherently encapsulate global information, we investigate patch tokens by incorporating a dedicated LPB for each layer. We first separate patch tokens from the complete token set as $\boldsymbol{T}_p$. Reshaping the patch embedding into a 2-D format allows for a detailed examination of their spatial characteristics. Leveraging the inherent local attention capability in convolutional operations, we conduct fine-grained learning on the spatial features of all segmented patches, aiming for more effective capture of small and densely distributed targets within the image. Here, $l \in [1, L]$ signifies the current encoder layer, and the operation unfolds as follows:

$$\boldsymbol{T}_p^l = \mathrm{Conv}\big(\mathrm{BN}\big(\mathrm{SiLU}\big(\mathrm{Conv}\big(\boldsymbol{T}_p^{l-1}\big)\big)\big)\big). \tag{1}$$

2) Global Perception Branch: The LPB provides the detailed regional context, while the conventional MHSA serves as the GPB, establishing long-range correlations among various tokens. Initially, the input token sequence undergoes normalization before being fed into the MHSA. Attention weights are computed by performing a scaled dot product between query (Q) and key (K), resulting in initial attention scores. The output tensor of each layer is derived via a linear transformation. It is worth noting that within our mechanism, both branches at each layer are combined and then proceeded to the subsequent layer, harmonizing fine-grained details and global information in the image. Within each layer, the input tokens $\boldsymbol{T} \in \mathbb{R}^{(C+N^2)\times D}$ are processed through the MHSA,

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resulting in layer-specific attention weights. Multihead attention enables the model to attend to diverse information across various representation subspace and positions. The formulation for this process is outlined as follows:

$$\boldsymbol{T}^l, \boldsymbol{A}^l = \mathrm{MHSA}\left(\boldsymbol{T}^{l-1}\right). \tag{2}$$

We combine two branches by adding them element-wisely to obtain the output tensor

$$\boldsymbol{T}^l = \boldsymbol{T}^{l-1}[C+1:] + \boldsymbol{T}_p^{l-1}. \tag{3}$$

3) Adaptive Attention Fusion: After passing through encoder layers, the final patch embedding is obtained and transformed into a 2-D tensor and then subjected to a dimension permutation, resulting in $\boldsymbol{T}_p^L \in \mathbb{R}^{D\times N\times N}$. Next, a convolutional layer is utilized to generate the initial $\mathrm{CAM}_{\mathrm{init}} \in \mathbb{R}^{C\times N\times N}$. Subsequently, through the adaptive attention fusion (AAF), we aggregate multihead attention weights from multiple layers to derive final class-specific attention weights and patch-level pairwise affinity weights. These weights are employed to refine the initial CAMs, enhancing their granularity.

Prior methods have commonly used a simple technique, averaging self-attention maps within each layer and summing them across layers, to merge representations from multiple attention heads. However, this mean-sum operation may dilute attention focus. Different attention heads typically emphasize distinct regions and classes, and the indiscriminate merging of attention heads introduces interference in the activation map for foreground objects. Therefore, we propose an adaptive fusion strategy to estimate the importance of different heads.

The self-attention maps $\boldsymbol{A} \in \mathbb{R}^{(L\times M)\times(C+N^2)\times(C+N^2)}$ represent attention across transformer layers, where M indicates the number of heads in each layer. Subsequently, we derive the dynamic weights $\boldsymbol{W} \in \mathbb{R}^{(M\times L)\times 1}$ via pooling across heads, followed by interaction within a feedforward neural network (FFN) network. Learning weights for each position allows the model to prioritize crucial heads, contributing to improved foreground activation. The specifics of these operations are outlined as follows:

$$\boldsymbol{W} = \mathrm{FFN}(\mathrm{Pooling}(\boldsymbol{A})) \tag{4}$$

$$\boldsymbol{A}' = \boldsymbol{A} \odot \boldsymbol{W}. \tag{5}$$

Utilizing element-wise multiplication, we derive the final enhanced attention denoted as $\boldsymbol{A}' \in \mathbb{R}^{(C+N^2)\times(C+N^2)}$. The attention matrix $\boldsymbol{A}'$ is split into class-specific attention weights $\boldsymbol{A}'_{\mathrm{c2p}} = \boldsymbol{A}'[1:C, C+1:C+N^2]$ and patch-level pairwise affinity weights $\boldsymbol{A}'_{\mathrm{p2p}} = \boldsymbol{A}'[C+1:C+N^2, C+1:C+N^2]$. Leveraging $\boldsymbol{A}'_{\mathrm{c2p}}$, we refine class-to-patch attention to capture image regions pertinent to each class. We further enhance CAMs by computing attention relationships between image patches through patch-level pairwise affinity weights. This enhancement emphasizes the interaction among image patches, refining CAMs' edges and details

$$\mathrm{CAM}_{\mathrm{c2p}} = \boldsymbol{A}'_{\mathrm{c2p}} \odot \mathrm{CAM}_{\mathrm{init}} \tag{6}$$

$$\mathrm{CAM}_f = \boldsymbol{A}'_{\mathrm{p2p}} \otimes \mathrm{CAM}_{\mathrm{c2p}} \tag{7}$$

where $\odot$ indicates the element-wise multiplication and $\otimes$ stands for matrix multiplication. After obtaining CAM_f, we proceed to generate initial pseudo masks.

C. Self-Supervised Consistency Constraint

In weakly supervised scenarios, supervision is limited, and the isotropic constraint, inherent in FS semantic segmentation, involves pixel-level labels undergoing similar spatial transformations as the input image during data augmentation. Nonetheless, this constraint diminishes when training CAMs using image-level supervision. To strengthen the supervision, we utilize a Siamese structure to generate two sets of inputs at different scales, providing mutual supervision for the stream without expanding the training dataset. This constraint aids the network in understanding features across multiple scales, rectifying CAM activation errors, and tackling the substantial challenges arising from diverse target scales in RS scenes.

Inspired by [8], we utilize a Siamese structure to implement equivariant consistency constraints on CAMs derived from images undergoing transformations, thereby offering self-supervised learning cues for pixel-wise classification. Initially, the input comprises an image pair $[I, I']$, and the backbone network extracts paired features, culminating in the generation of the final two streams CAM_f. Using bilinear interpolation, the CAMs undergo resizing to conform to the input size while simultaneously mapping their values into [0, 1]. Then, a distance loss is calculated between two CAMs, aiming to minimize disparities at the pixel level

$$L_{\mathrm{cc}} = \sqrt{\sum_{i=1}^{H}\sum_{j=1}^{W}\left\|\left(\mathrm{CAM}_f\left(x_i, y_j\right) - \left(\mathrm{CAM}'_f\left(x_i, y_j\right)\right\|\right.}. \tag{8}$$

Apart from introducing the self-supervised consistency constraint, we also consider the interplay between two streams when formulating additional loss functions, and the details are elaborated in Section III-E.

D. Task-Driven Joint Optimization

Using image-level supervision to generate CAMs often results in unclear object boundaries across the entire region and activation errors. Postprocessing techniques are commonly applied in WSSS to refine initial pseudo masks, such as methods based on pixel affinities or additional saliency map optimization. IRNet [47] identifies confident seed regions of the target class from attention maps of a classification model and propagates these seeds to determine accurate boundaries for entire instance regions. AffinityNet [9] employs a local response mechanism to fine-tune CAMs by disseminating information to neighboring regions with shared semantic entities, predicting the semantic connections between adjacent image coordinates. While these methods have notably improved pseudo masks, they necessitate training a separate network, often demanding substantial computational resources and hyperparameter tuning.

Instead of employing conventional and intricate retraining and optimization steps, we develop a unified optimization strategy tailored specifically for RS scenarios. This approach harnesses the potential of a generic foundational model to tackle issues concerning unclear object boundaries and errors in semantic distribution, thereby improving the quality of generated pseudo masks. SAM, a segmentation foundational model trained on over 1.1 billion images with 1 billion masks,

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Algorithm 1 JOS

Require:
Initial Pseudo Mask P_m, SAM Segments S_{sam}, Class size C, Overlap Threshold θ.
RSK_1: The area of segment should be within certain range.
RSK_2: Each class segment should possess certain low-level visual cues.

Ensure: Final Pseudo Mask P_f

procedure JOS(P_m, S_{sam})
  for seg**in**S_{sam} **do**
    if seg not satisfies RSK_1 **then**
      continue
    else
      $P_f \leftarrow$ np.zeros_like(P_m)
      for i **in** [1, C] **do**
        $P_c = (P_m == i)$
        if P_c **is** all False **then**
          continue
        else$(P_c \cap seg)! = 0$
          $lap_p = (P_c \cap seg)/P_c$
          $lap_s = (P_c \cap seg)/seg$
          if $lap_p >= \theta$ *or* $lap_s >= \theta$ **then**
            if $lap_p >= lap_s$ **then**
              $Cand = P_c$
            else $lap_p < lap_s$
              $Cand = seg$
            end if
            if $Cand$ satisfy RSK_2 **then**
              $P_f[Cand$ is True] = i
            end if
          else
            continue
          end if
        end if
      end for
    end if
  end for
  return P_f
end procedure

demonstrates remarkable zero-shot capability in segmenting images into semantically meaningful yet class-agnostic regions. Our approach enables achieving performance similar to the pseudo masks generated by complex and computationally intensive postprocessing modules, utilizing fewer computational resources. Furthermore, it seamlessly integrates into existing WSSS models without necessitating modifications to the underlying network structure.

The key steps include identifying overlaps between each class's pseudo mask and segments, calculating their respective overlap ratios, and incorporating segments that surpass certain threshold θ into the new pseudo mask for that class. If no overlap is found, we retain the original mask. Furthermore, leveraging domain-specific knowledge in RS scenarios, we refine the pseudo masks by referring to the visual cues of respective categories. For instance, certain categories in RS showcase unique visual attributes. Entities with these attributes, such as storage tanks (STs) known for their elliptical shapes, bridges recognized for their slender structures and aspect ratios, and ships commonly appear in proximity to harbors (i.e. RSK2 in Algorithm 1). Techniques, such as geometric detection and relation reasoning, enable the extraction of nuanced visual cues for these entities, which serves for optimizing the pseudo mask to mitigate the category confusion. When deviations from these features are detected, the corresponding pseudo masks undergo automatic filtering. The detailed process is described in Algorithm 1.

E. Loss Functions

Besides the class consistency constraint introduced in Section III-C, this section offers an elaborate elucidation of additional losses pertinent to our model. The classification loss is primarily used for supervised learning from weakly labeled image data to guide the network in learning correct semantic categories. The consistency constraint emphasizes the model's consistency across different scales and views, contributing to improved generalization performance. The contrastive loss focuses on the similarity between class labels, alleviating interference among classes. The final loss $L_{\sum}$ is the weighted sum of three components: classification loss (cls), consistency constraint (cc), and contrastive loss (ct)

$$L_{\sum} = \lambda_1 L_{\text{cls}} + \lambda_2 L_{\text{cc}} + \lambda_3 L_{\text{ct}}. \tag{9}$$

1) Classification Loss: During the training phase, for image-level labels, most of the relevant research in target localization uses the fundamental multilabel CE loss [52]. Within weakly supervised settings, CAM is applied for average pooling to acquire the corresponding class scores. In our approach, we compute class scores using both class tokens and patch tokens, balancing the classification loss by averaging scores from patch tokens and the final CAM_f via average pooling. Specifically, we segment the final output tokens $\boldsymbol{T}^L$ from the encoder into two categories: class tokens ($\boldsymbol{T}_c^L \in R^{C\times D}$) undergo the averaging to compute probabilities for each category, while patch tokens ($\boldsymbol{T}_p^L \in R^{N^2\times D}$) reshaped to a spatial tensor and resulted in C channels via convolution and pooling operations. We integrate CAM_f to derive the final category scores for predictions as follows:

$$L_{\text{cls}} = L_c + L_p + L_f \tag{10}$$

where L_c, L_p, and L_f, respectively, represent the losses of the class token, patch token, and final CAM components.

In RS scenarios, class imbalance in data is pronounced, leading networks to disregard classes with lower frequencies. To mitigate this, we adopt an asymmetric loss [53] format. This format assigns the current class as the positive sample and treats all other classes as negative ones. By dynamically adjusting the weight of less challenging negative samples using the threshold value $m = 0.01$, it diminishes the overall contribution of negative samples to the loss function. The asymmetric loss replaces the conventional CE loss as follows:

$$L_{\text{al}} = -y(1-p)^{\gamma^+}\log(p) - (1-y)p_m^{\gamma^-}\log(1-p_m) \tag{11}$$

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where p denotes the predicted probability for the corresponding class, y signifies the true class label, and γ acts as a hyperparameter suppressing easier samples with a higher effect on negative ones, with $\gamma^+ = 0$, $\gamma^- = 2$. Within our network, two sets of scores, y and y', are computed for the original image $\boldsymbol{I}$ and the transformed image $\boldsymbol{I}'$. The overall classification loss is formulated as

$$L_{\text{cls}} = \frac{1}{2} L_{\text{cls}}(y, \boldsymbol{I}) + \frac{1}{2} L_{\text{cls}}(y', \boldsymbol{I}'). \tag{12}$$

2) Contrastive Loss: The classification loss enables distinct class tokens to emphasize various object regions, yet the resulting attention maps from diverse class tokens to image regions commonly feature overlapping target areas within the same image. This occurrence arises due to two factors: 1) employing a multilabel one-to-many loss for computing class scores, amalgamating embeddings of each class token, which does not rigorously ensure differentiation among distinct class tokens; and 2) confining the loss application solely to the output class tokens of the final layer, thereby limiting its impact mostly to the final layer of the network. Intermediate or lower layers might lack adequate guidance to effectively discern different class tokens.

To mitigate class interference, a concatenation operation is applied to every obtained class token at each layer. Normalizing the class token $\boldsymbol{T}_c \in \mathbb{R}^{M \times C \times N^2}$ along the last dimension, we derive a similarity score matrix by performing matrix multiplication between $\boldsymbol{T}_c$ and its transposed tensor. To enforce each class token's self-similarity, we calculate the CE loss between the similarity matrix $\boldsymbol{S} \in \mathbb{R}^{M \times C \times C}$ and repeated ground-truth tensor $\boldsymbol{Y} \in \mathbb{R}^{M \times C}$. This strategy enhances the learning of more discriminative class-specific features across different network layers, encompassing both high-level semantic and low-level spatial features. The specific formula is as follows:

$$\boldsymbol{S}^l = \boldsymbol{T}_c^l \cdot \left(\boldsymbol{T}_c^l\right)^T \tag{13}$$

$$L_{\text{ct}} = \frac{1}{L} \sum_{i=1}^{L} \text{CELoss}(\boldsymbol{S}^l, \boldsymbol{Y}). \tag{14}$$

IV. Experiments

For our experimental evaluation, we comprehensively consider a public large-scale dataset focusing on multicategory airborne image segmentation and the common dataset focusing on urban and rural buildings: iSAID [54], ISPRS Potsdam,[1] and ISPRS Vaihingen.[2] These three datasets substantiate the competitiveness of our approach from various perspectives. We initiate by analyzing the data to understand the distribution characteristics and challenges across datasets. Following this, we present a comparative analysis against SOTA models, accompanied by visualizations. In addition, we showcase the results of ablation experiments, confirming the efficacy of each module in our model. Finally, we offer a detailed analysis of hyperparameter sensitivity.

[1]https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-potsdam.aspx

[2]https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-vaihingen.aspx

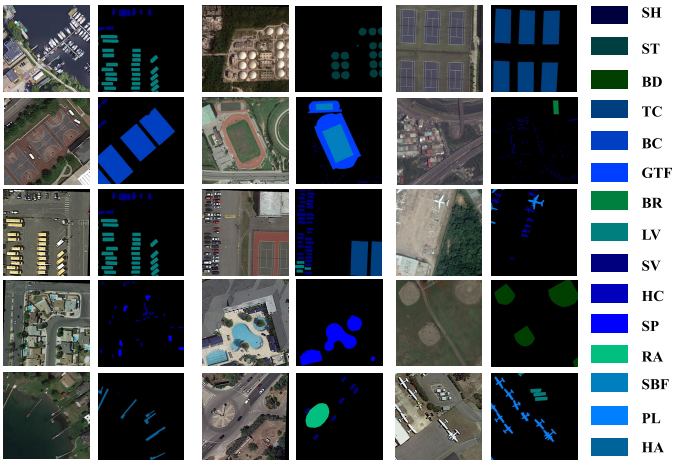


Fig. 4. Example images and ground-truth masks in the iSAID dataset.

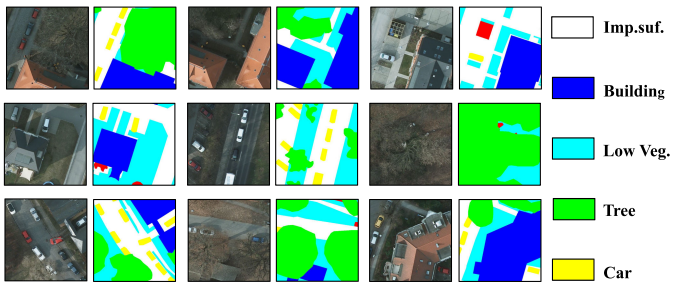


Fig. 5. Example images and ground-truth masks in the Potsdam dataset.

A. Dataset and Evaluation Metrics

1) iSAID: iSAID is a publicly available aerial image segmentation dataset, mainly captured by JL-1 and GF-2 satellites, with the majority of images obtained from Google Earth. It contains numerous instances with diverse scale variations. The dataset comprises 15 categories and one background category (BG), encompassing ship (SH), ST, baseball diamond (BD), tennis court (TC), basketball court (BC), ground track field (GTF), bridge (BR), large vehicle (LV), small vehicle (SV), helicopter (HC), swimming pool (SP), roundabout (RA), soccer ball field (SBF), plane (PL), and harbor (HA). Please refer to Fig. 4 for sample images displaying all categories alongside their pixel-wise annotations. The dataset originally consists of 1411 high-resolution images designated for training and validation, with the remaining 458 images designated for testing. Following the guidelines in previous works [12], we randomly extract 6–8 slices sized 512 × 512 from each original image. These slices are intentionally selected based on the distribution of foreground positions to better simulate real-world scenarios. Consequently, our experiments employ approximately 8700 training slices and 2800 validation slices as the final image sets.

2) ISPRS: The ISPRS dataset comprises aerial images portraying two distinct scenes: Potsdam, a traditional city exhibiting tall buildings and narrow streets, and Vaihingen, a small village characterized by numerous independent houses. Both datasets exhibit consistent category types and high similarity. We specifically focused on the analysis of the ISPRS Potsdam dataset. In addition, to further validate our method,

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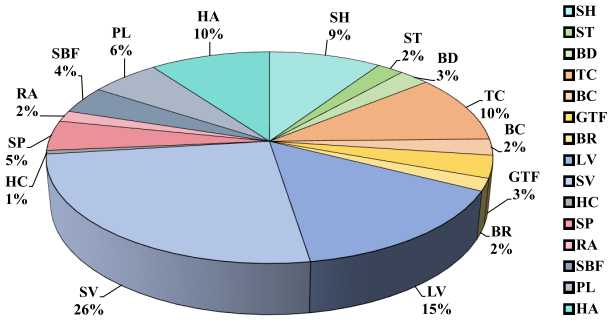


Fig. 6. Percentage of images for respective categories in the iSAID dataset.

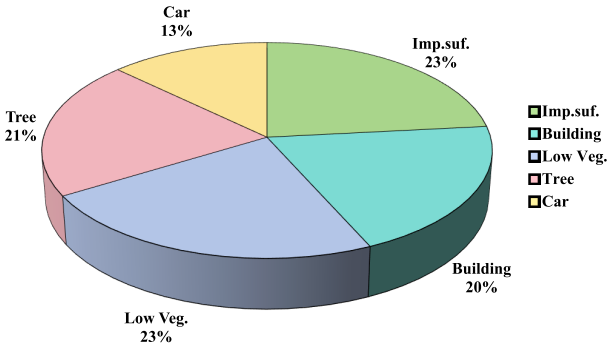


Fig. 7. Percentage of images for respective categories in the Potsdam dataset.

we conducted experiments on the Vaihingen dataset. The ISPRS Potsdam dataset comprises 37 high-resolution images at a pixel resolution of 0.5 m, each image with 6000 × 6000 pixels. These images are split into 23 and 14 for training and testing, respectively. After removing clutter categories, there are five land cover types: impervious surfaces (Imp.suf.), building, low vegetation (Low Veg.), tree, and car. To facilitate augmentation, images are cropped into 512 × 512 pixels with a 256-pixel overlap. The cropped images showcasing all categories with ground-truth annotations are shown in Fig. 5. Similarly, according to the official benchmark, in the ISPRS Vaihingen dataset, we specify the use of 16 images for model training and 17 images for testing. The images were segmented into 256 × 256, with a 128-pixel overlap.

For fair comparisons against SOTA models, we keep the experimental protocols consistent across datasets and refer to their open-sourced implementations for generating segmentation results.

3) Class Distribution: As shown in Fig. 6, the iSAID dataset's category distribution illustrates the quantity of each category post-partitioning of the original training dataset, maintaining alignment with its initial distribution. However, a notable imbalance prevails, with LV and SV being more prevalent, while HC, BR, and RA occur less frequently. Experimental results highlight the insufficient amount of training samples in these categories, impeding effective feature learning and leading to relatively lower accuracy. The Potsdam dataset exhibits a relatively balanced category distribution in both the original and partitioned datasets, as depicted in Fig. 7. The car category shows a relatively lower quantity, while the distribution of other categories remains consistently stable.

4) Class Co-Occurrence: We further examine the co-occurrence of categories within the same image, which aids in the subsequent analysis of category dependency and distinguishability. The matrix visualized in Fig. 8 illustrates the co-occurrence patterns within the iSAID dataset. Specifically, the SH consistently accompanies the presence of the HA, and the GTF is often associated with the SBF, while the SV tends to be observed with the appearance of the LV. The visual analysis of co-occurrence aids in devising optimization strategies informed by expert knowledge. Given the limited occurrence of the car category in the Potsdam dataset, its values in the co-occurrence matrix notably diminish. Fig. 9 illustrates the observable co-occurrence between the car and the Imp.suf. Meanwhile, for the building category, there is a notable co-occurrence pattern with all other categories.

5) Evaluation Metrics: Utilizing the above two sets, we perform a thorough quantitative assessment of our model, employing metrics, including average intersection over union (IoU), F_1 score, and overall accuracy (OA). Adhering to established standards [11], we adopt true positive (TP), true negative (TN), false positive (FP), and false negative (FN) to denote these metrics. The definitions of mIoU, F_1 score, and OA are as follows:

$$\text{mIoU} = \frac{1}{C}\sum_{i=1}^{C}\frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \tag{15}$$

$$\text{OA} = \frac{\text{TP}+\text{TN}}{\text{TP}+\text{TN}+\text{FP}+\text{FN}} \tag{16}$$

$$F_1 = \frac{2 \odot P \odot R}{P+R} \tag{17}$$

$$\text{mF1} = \frac{1}{C}\sum_{i=1}^{C} F_{1i}. \tag{18}$$

Precision (P) is computed as $P = \text{TP}/(\text{TP}+\text{FP})$ and recall (R) is computed as $R = \text{TP}/(\text{TP}+\text{FN})$, where C represents the number of categories. mF1 evaluates the classification task performance. IoU assesses segmentation model accuracy by measuring the IoU ratio between prediction and ground truth. mIoU averages IoU across all categories, offering a comprehensive assessment of overall performance. OA evaluates the overall accuracy of the classification model by representing the ratio of correctly classified pixels to the total pixels in the sample set. The F_1 score, a composite metric of precision and recall, ranges from 0 to 1.

B. Implementation Details

Experiments are conducted on Ubuntu 20.04, leveraging the NVIDIA GeForce RTX 3090 and CUDA 11.7 environment. During the training phase, we define 60 epochs for training with a batch size of 96. Using a cosine annealing learning rate strategy, we initialize the learning rate to $5e^{-4}$, scaling it based on the batch size: $\text{lrscale} = \text{lr}/(512 \times \text{batchsize})$. Furthermore, we initiate a warm-up learning rate for the initial five epochs, beginning at a default of $1e^{-6}$. Data augmentations align with the settings in the Deit [55] framework. In the inference phase, a multiscale approach is used to generate the final CAMs by merging CAMs of varying scales. Next, a threshold is applied to filter the background, resulting in the initial pseudo mask. The pseudo mask undergoes additional

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TABLE I
COMPARISON OF GENERATED PSEUDO MASKS ON THE iSAID, MEASURED IN mIoU (%)

Method	BG	SH	ST	BD	TC	BC	GTF	BR	LV	SV	HC	SP	RA	SBF	PL	HA	mIoU
Affinitynet	51.52	*25.01*	*23.75*	16.45	39.21	*33.93*	*14.67*	7.35	*21.19*	3.54	0.10	2.32	*8.74*	*46.20*	22.72	7.31	20.22
Irnet	87.46	6.18	18.31	20.82	9.57	5.26	10.25	3.55	7.52	8.75	0.10	4.99	6.70	3.21	18.11	2.40	14.38
SEAM	*89.50*	8.58	16.57	15.78	59.03	2.59	6.71	6.21	14.76	3.77	0.25	2.31	5.09	11.94	10.04	1.56	15.85
Advcam	87.76	7.43	20.95	23.55	19.10	6.47	8.78	2.91	10.06	*8.76*	0.30	5.19	5.66	3.77	21.88	1.97	14.75
Mctformer	81.59	15.97	15.64	*41.58*	45.61	26.70	11.33	*9.83*	10.82	4.83	9.73	*22.65*	5.95	11.36	40.52	9.56	22.72
WeakTr	82.40	15.22	19.62	37.28	*60.10*	26.45	13.82	6.48	15.56	5.38	*11.78*	16.92	4.25	20.73	*48.62*	*15.51*	*25.06*
Ours	**90.10**	**35.43**	**47.10**	**64.37**	**70.10**	**60.17**	**38.10**	**26.55**	**41.10**	**29.34**	**22.17**	**45.31**	**26.34**	**48.37**	**57.13**	**33.27**	**45.93**

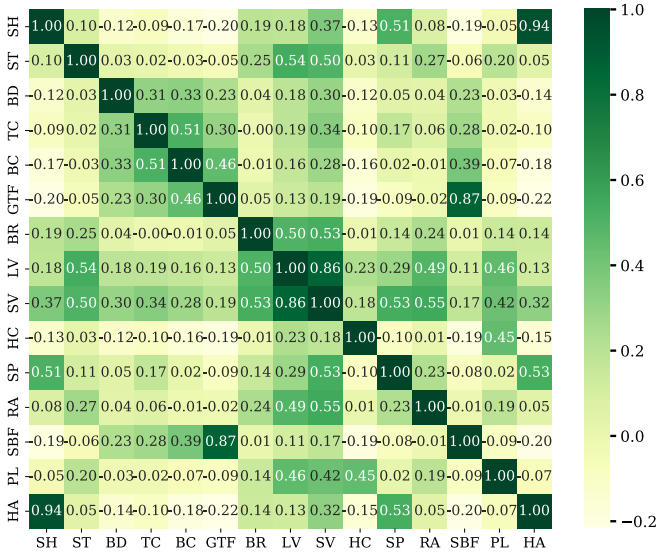


Fig. 8. Class co-occurrence matrix of iSAID.



Fig. 9. Class co-occurrence matrix of Potsdam.

refinement using denseCRF, followed by our designed JOS to generate the final mask. For the segmentation training phase, we maintain the consistency of existing works [12] by employing the DeepLabV1 network, adhering to principles for fair comparisons.

C. Comparison Methods

We evaluate six SOTA WSSS methods in comparison to our proposed approach, which are summarized as follows.

1) *Affinitynet [9]:* CAMs typically focus on delineating locally distinct elements. AffinityNet accomplishes semantic propagation through a random walk strategy, spreading local responses to nearby regions that share the same semantic entity.
2) *IRNet:* Ahn et al. [47] identify confident seed regions of the target class from attention maps generated by a classification model, subsequently propagating these seeds to cover entire instance regions with accurate boundaries.
3) *Advcam:* Lee et al. [56] employ an anti-adversarial approach, perturbing images along pixel gradients contrary to those used in adversarial attacks. This manipulation compels initially considered nondiscriminative regions to contribute to subsequent classifications, generating CAMs that progressively identify expanded regions of the target object.
4) *SEAM [8]:* It proposes a consistency regularization approach for CAMs predictions. By applying equivariance constraints to CAMs of augmented images, it provides a self-supervision signal for training.
5) *Mctformer (Baseline):* Xu et al. [7] investigate whether a transformer model can effectively capture class-specific attention for more discriminative object localization by incorporating multiple class tokens within the transformer architecture. This method is also served as our baseline.
6) *Weaktr [57]:* In the context of noisy label issues, studies suggest that samples with smaller gradients are often considered cleaner. WeakTr investigates a gradient clipping decoder based on ViT, determining whether to clip the gradient at each pixel via thresholding.

D. Experimental Results

1) iSAID Dataset: This section assesses the performance of our CocoaNet model on the iSAID dataset across three facets: comparisons with SOTA methods, performance gaps with FS counterparts, and qualitative result analysis through visualization.

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TABLE II
COMPARISON OF SEGMENTATION RESULTS ON THE iSAID, MEASURED IN TERMS OF mIoU (%), OA (%), AND F_1 (%)

Method	BG	SH	ST	BD	TC	BC	GTF	BR	LV	SV	HC	SP	RA	SBF	PL	HA	mIoU	mF1	OA
Affinitynet	58.08	*28.59*	19.84	22.45	39.43	*40.63*	9.01	7.39	*33.73*	3.33	0.47	2.96	5.47	*60.74*	33.02	8.41	23.32	34.14	85.37
Irnet	86.45	1.01	8.53	21.76	5.96	2.76	3.84	1.35	1.48	1.58	0.32	1.39	6.70	1.78	7.41	1.41	14.38	11.85	65.34
SEAM	88.13	2.37	6.57	14.72	61.36	1.31	1.11	3.21	2.88	1.31	0.59	0.41	1.94	1.44	5.21	1.01	13.09	12.48	67.32
Advcam	86.67	0.27	10.88	21.31	11.37	3.51	8.78	1.91	2.06	2.76	0.51	3.19	2.66	2.77	8.88	1.97	14.75	13.61	68.31
Mctformer	76.63	16.10	*41.10*	39.98	35.70	39.83	10.40	7.51	17.11	2.79	*3.31*	34.45	*15.66*	35.67	*55.31*	10.31	27.61	39.54	88.21
WeakTr	*88.65*	16.53	26.53	*52.26*	*67.94*	35.51	*12.47*	*18.76*	29.95	*7.59*	1.31	*48.49*	1.35	16.62	51.57	*10.88*	*30.40*	*41.52*	*93.51*
Ours	**92.11**	**38.10**	**56.10**	**69.31**	**71.23**	**59.10**	**40.41**	**22.31**	**53.31**	**29.13**	**13.21**	**49.37**	**31.43**	**64.67**	**63.55**	**34.17**	**49.22**	**63.47**	**97.13**

TABLE III
PERFORMANCE GAP WITH FS COUNTERPART ON THE iSAID, MEASURED IN TERMS OF mIoU (%), OA (%), AND F_1 (%)

Supervision	BG	SH	ST	BD	TC	BC	GTF	BR	LV	SV	HC	SP	RA	SBF	PL	HA	mIoU	mF1	OA
Our Masks	92.11	38.10	56.10	69.31	71.23	59.10	40.41	22.31	53.31	29.13	13.21	49.37	31.43	64.67	63.55	34.17	49.22	63.47	97.13
Ground Truth	95.22	45.20	64.87	79.17	89.27	67.25	51.70	22.70	60.88	34.96	14.67	51.44	69.16	76.39	69.26	41.26	58.27	69.03	99.42

a) Comparisons with SOTAs: As demonstrated in Table I, our initial assessment focuses on comparing pseudo mask quality among various methods. Our method significantly enhances mask quality, particularly for large-scale targets such as BC and SBF, exhibiting improvements of 33.47% and 37.01% over the baseline (Mctformer), highlighting the effectiveness of our two-stream constraint. In addition, the integration of an LPB leads to improvements of 19.46% and 24.51% for smaller and denser targets such as SH and SV, relative to the baseline. The mIoU quality of our masks notably exceeds other methods by 23.21% over the baseline, providing more precise masks for subsequent segmentation models. We present the segmentation results in Table II, utilizing respective pseudo masks for supervision. These results reveal substantial enhancements across all categories, effectively mitigating background interference. In particular, BD, TC, LV, SV, and SBF exhibit improvements exceeding 30% compared to the baseline. Despite the HC category's relatively small sample size, our model demonstrates exceptional performance. This further validates the effectiveness of our method, representing significant advancements in WSSS.

b) Gaps with FS model: Table III presents the comparison between the final segmentation accuracy of our proposed model and FS counterparts on the iSAID dataset. Utilizing the Deeplabv1 under the supervision of our generated masks, we match the FS models in categories such as BR, SV, and SP. However, there still exist considerable gaps in certain categories, such as GTF and RA, closely associated with their intrinsic features. The visual patterns of GTF are relatively less distinct, while RA typically possesses diverse textures and colors, posing challenges for the training under image-level class labels.

c) Visualization: In Fig. 10, we illustrate the qualitative comparisons of CAM visualization between our model and others on the iSAID dataset. Clearly, our approach not only delivers more accurate fine-grained information but also significantly reduces the number of incorrectly activated pixels in the background, resulting in sharper edges for the targets. It can be observed that our approach achieves more accurate segmentation in densely distributed LV, PL, SH, and ST, rectifying underactivation issues present in other methods. Larger targets, such as SBF, TC, HA, and BD, exhibit clearer boundaries and regular shapes in our CAM visualizations. Fig. 11 depicts the segmentation comparisons on the iSAID dataset. Our model achieves clearer segmentation of dense small targets and establishes well-defined boundaries for large-scale targets as highlighted in red boxes. For example, our model exhibits significantly improved performance compared to the baseline for GTF and SBF categories. We accurately identify their boundaries and delineate complete target regions. For images with multiclass targets present simultaneously, such as SH, TC, GTF, and SBF, our method manages to distinguish them, while the baseline shows overlapping boundaries.

2) Potsdam Dataset: In addition to the iSAID dataset, we extend our method to building a detection dataset, ISPRS Potsdam, to further validate our approach.

a) Comparisons with SOTAs: The Potsdam dataset contains fewer categories compared to iSAID. The co-occurrence between classes is more frequent, such as Low Veg. that often appears alongside tree. Some categories, such as Imp.suf. and Low Veg., lack distinct boundaries. Despite these challenges, our model generates pseudo masks with higher quality. For instance, from Table IV, categories, such as building and tree, display improvements of 32.43% and 51.37%, respectively, compared to the baseline. Even the relatively small category of cars demonstrates a 28.54% improvement. The mIoU measure for pseudo mask quality rises by 35.28% compared to the baseline, ensuring mask quality for subsequent segmentation.

The segmentation results, detailed in Table V, demonstrate significant enhancements across all categories compared to the baseline. Specifically, there are notable improvements in IoU scores for Imp.suf., building, Low Veg., tree, and car, showing increases of 56.47%, 31.99%, 28.63%, 61.06%, and 31.23%, respectively. While Irnet displays higher IoU results for Imp.suf., its final segmentation results are 8.52% lower

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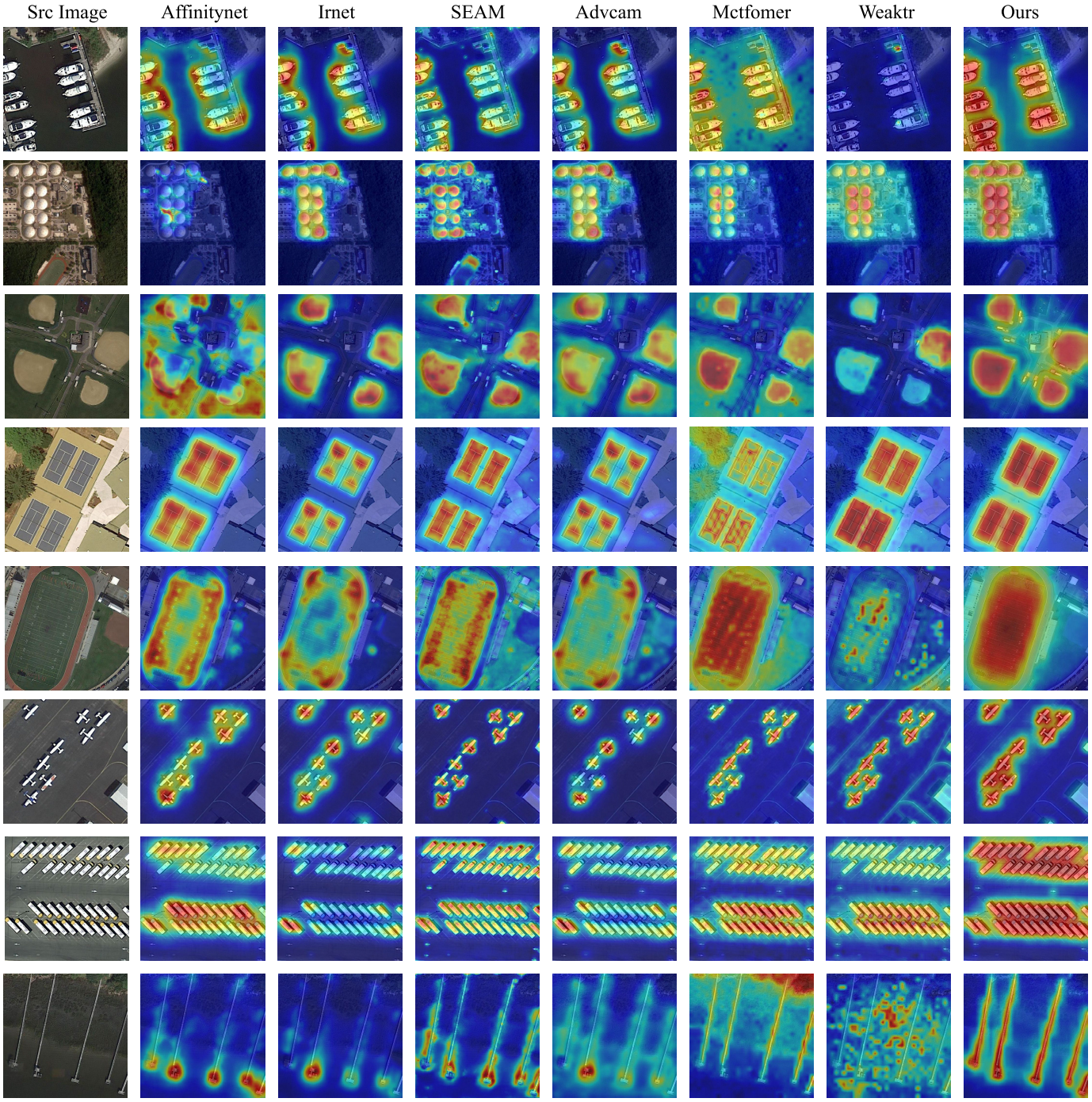


Fig. 10. Qualitative comparison of CAM results obtained by different methods on the iSAID dataset.

than our model, indicating the superior quality of our pseudo masks. Our model attains an OA of 95.31%, and the mIoU significantly surpasses other methods, further validating its overall effectiveness.

b) Gaps with FS model: Table VI illustrates the final segmentation accuracy of the proposed WSSS against FS model on the Potsdam. While our model nearly matches FS results in categories such as building and tree, disparities persist in the Imp.suf. and Low Veg. These categories suffer due to ambiguous boundaries and lack of distinct visual features, resulting in inferior pseudo mask quality. The coexistence of car and Imp.suf. categories also introduces complexities in model training. Despite these disadvantages, our model achieves a substantial comparability against fully supervised semantic segmentation (FSSS) in overall performance.

c) Visualization: Fig. 12 presents the qualitative comparisons on the Potsdam dataset, demonstrating that our approach generates clearer CAMs without misactivation issues compared to the baseline. Especially in the Imp.suf. category, where the presence through buildings and less distinct features challenge segmentation, our method accurately delineates the corresponding regions and exhibits clear boundaries. Fig. 13 demonstrates substantial enhancement in the final segmentation results compared to the baseline. Specifically,

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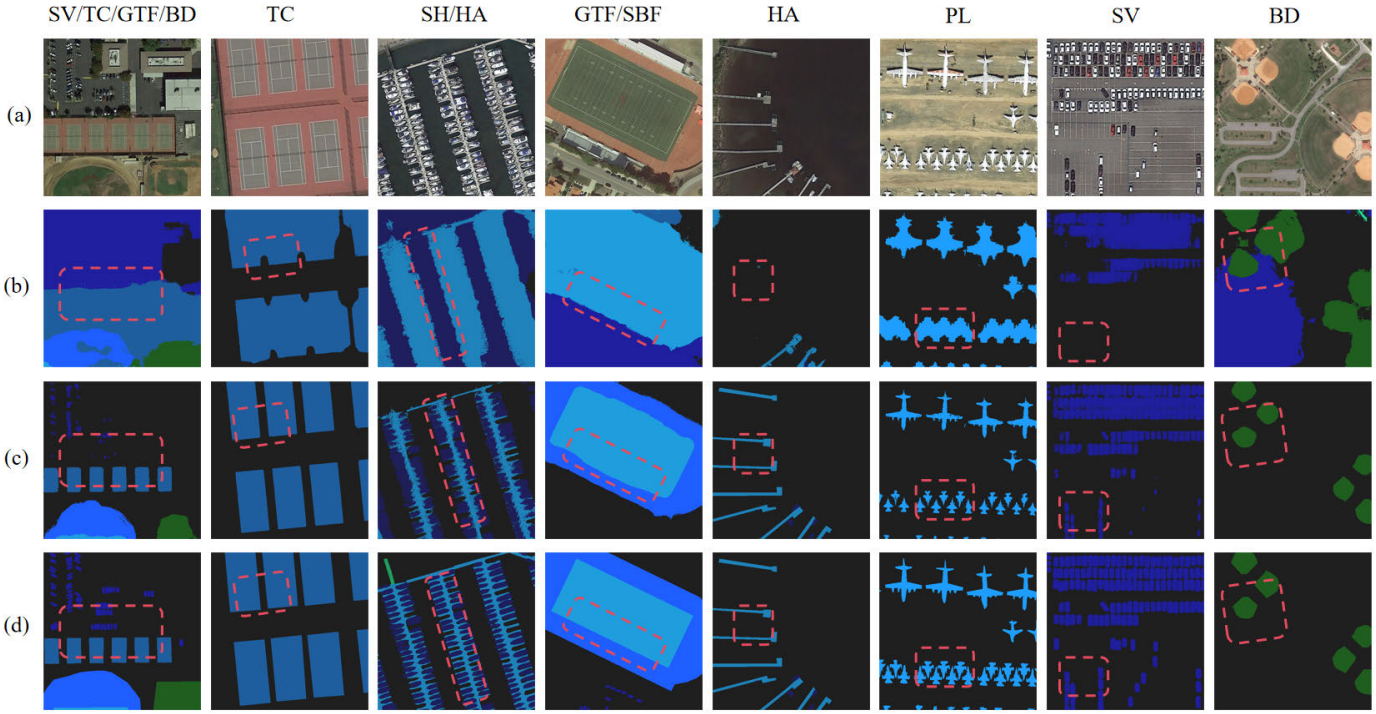


Fig. 11. Visualization of segmentation results on the iSAID dataset. (a) Image. (b) Baseline. (c) Ours. (d) Ground truth. We present visual results of various-sized RS targets, including SV, TC, SH, SBF, PL, BD, and HA.

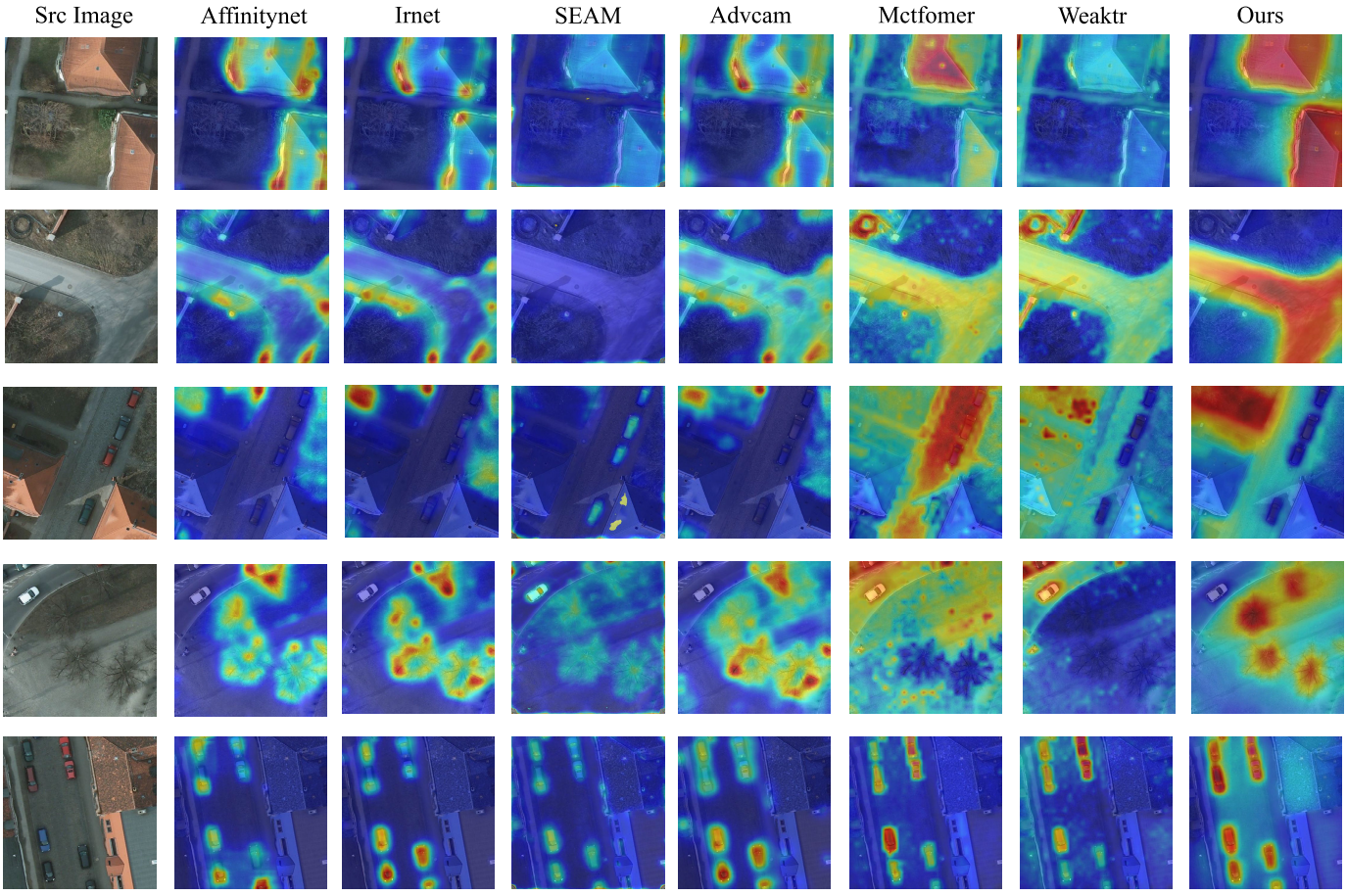


Fig. 12. Qualitative comparison of CAM results obtained by various methods on the Potsdam dataset.

our method accurately identifies the Imp.suf. category, while the baseline results in confusion and irregular boundaries. In scenarios where multiple categories coexist within the same area, such as Low Veg. and car sharing the same space, our model successfully delineates clear boundaries for each category.

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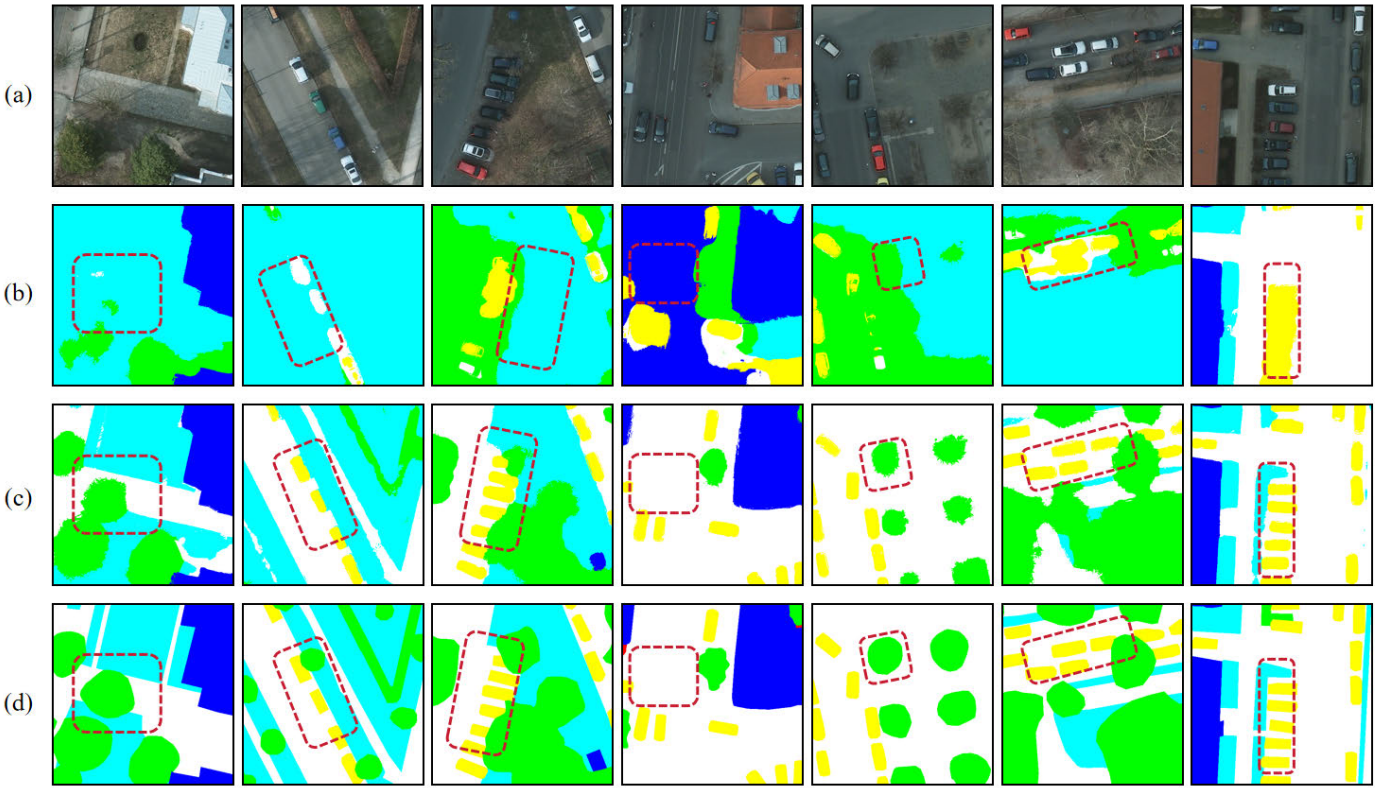


Fig. 13. Visualization of segmentation results on the Potsdam dataset. (a) Image. (b) Baseline. (c) Ours. (d) Ground truth.

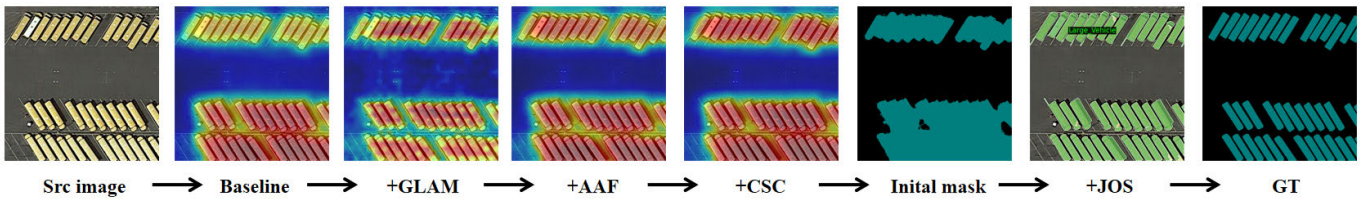


Fig. 14. Visualization results of CAMs on the iSAID dataset after adding different modules to the baseline. The addition of the GLAM module improves target localization, while the AAF and CSC modules enhance the granularity of CAM. Building upon these three modules, the JOS strategy is employed to optimize the pseudo mask, thereby better meeting the requirements of segmentation masks.

TABLE IV
COMPARISON OF GENERATED PSEUDO MASKS ON THE POTSDAM, MEASURED IN MIOU (%)

Method	Imp.suf.	Building	Low Veg.	Tree	Car	mIoU
Affinitynet	27.12	*61.44*	*49.61*	16.24	1.80	31.24
Irnet	**47.41**	53.74	48.76	28.19	19.43	39.51
SEAM	11.61	49.21	15.31	32.41	6.62	23.03
Advcam	*39.06*	56.43	46.58	*47.88*	*31.78*	*44.35*
Mctformer	13.17	38.38	24.42	15.95	14.57	21.30
WeakTr	16.11	46.48	28.52	21.76	14.43	25.46
Ours	45.31	**70.81**	**56.58**	**67.32**	**43.11**	**56.40**

3) Vaihingen Dataset: To further validate the reliability of our approach, we conduct additional experiments on the Vaihingen dataset.

Table VII shows that our method performs well across all categories, demonstrating the quality of our pseudo masks. There is a significant improvement in each category compared to the baseline. For example, the Imp.suf. category has increased by 36.31%, the car category has improved by 37.88%, and the mIoU has shown an increase of 33.44%. Table VIII shows that our method yields favorable segmentation results across all categories, confirming the effectiveness of our approach. For the Imp.suf. category, where the Affinitynet method achieves the highest precision in initial CAM in Table VII, our method outperforms compared methods in this category for the final segmentation in Table VIII, further substantiating the efficacy of our pseudo mask guidance. Our metrics far surpass those of other methods in the car category, demonstrating the capability of capturing small targets. Compared to the baseline, improvements are 29.41% in mIoU and 24.07% in mF1.

E. Ablation Experiments

In this section, we systematically discuss the effectiveness of the proposed main components: global–local attention mechanism (GLAM), self-supervised consistency constraint (CSC),

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TABLE V
COMPARISON OF SEGMENTATION RESULTS ON THE POTSDAM, MEASURED IN TERMS OF MIOU (%), OA (%), AND F_1 (%)

Method	Imp.suf.	Building	Low Veg.	Tree	Car	mIoU	mF1	OA
Affinitynet	29.01	*71.67*	44.67	7.35	1.80	31.45	40.78	86.22
Irnet	*61.19*	62.90	48.02	21.61	1.04	38.95	50.73	90.32
SEAM	12.73	63.81	10.31	38.63	7.51	26.61	38.76	84.40
Advcam	53.53	60.78	*52.56*	*58.51*	34.24	*51.92*	*67.81*	*91.24*
Mctformer	13.24	57.32	26.84	4.25	16.23	23.58	31.51	83.07
WeakTr	17.28	68.75	32.20	21.35	*35.09*	34.93	46.18	85.36
Ours	**69.71**	**89.31**	**55.47**	**65.31**	**57.55**	**67.47**	**78.31**	**95.31**

TABLE VI
PERFORMANCE GAP WITH FS COUNTERPART ON THE POTSDAM, MEASURED IN TERMS OF MIOU (%), OA (%), AND F_1 (%)

Supervision	Imp.suf.	Building	Low Veg.	Tree	Car	mIoU	mF1	OA
Our Masks	69.71	89.31	55.47	65.31	57.55	67.47	78.31	95.31
Ground Truth	85.56	93.03	75.26	73.79	78.28	81.19	89.45	96.94

TABLE VII
COMPARISON OF GENERATED PSEUDO MASKS ON THE VAIHINGEN, MEASURED IN MIOU (%)

Method	Imp.suf.	Building	Low Veg.	Tree	Car	mIoU
Affinitynet	**62.91**	*62.13*	*35.62*	*53.97*	2.98	*43.45*
Irnet	18.22	18.58	10.61	13.92	*22.73*	16.81
SEAM	37.44	36.35	12.31	38.23	4.18	25.71
Advcam	8.46	5.28	11.01	8.06	2.07	6.98
Mctformer	25.72	39.64	26.01	46.27	7.28	28.99
WeakTr	25.11	42.46	25.72	39.04	7.41	27.86
Ours	*62.03*	**74.87**	**65.79**	**64.32**	**45.16**	**62.43**

task-driven JOS, and improved loss function. All experimental and visual results presented are based on the iSAID dataset.

1) Main Components: For a more rigorous evaluation in our ablation experiments, we select Mctformer as the baseline and utilize the mIoU of pseudo masks as the evaluation metric to assess each component's effectiveness in the framework. To illustrate our approach's effectiveness, we sequentially integrate the GLAM, AAF, CSC, CRF, and JOS modules into the baseline. The detailed results are provided in Table IX. Our network has significantly improved overall compared to the baseline. The introduction of the GLAM in the single stream significantly boosts performance across all aspects. It specifically enhances the handling of dense small targets, with LV and SV showing improvements of 12.50% and 6.32%, respectively. Moreover, for large-scale targets such as BC and SBF, the enhancements reach 18.58% and 30.60%, respectively. The blurry nature of BC and the co-occurrence between SBF and GTF contribute to the poor quality of pseudo masks in the baseline. However, the introduction of our attention mechanism significantly improves local attention capacity, refining the granularity of CAMs. In addition, the incorporation of AAF helps focus attention on target regions, resulting in substantial enhancements across all categories and an overall mIoU increase of 3.13%. Addressing the significant scale variations of targets across scenes, we implement a two-stream architecture to control learned target features for diverse image sizes. Consequently, the adaptation of CSC results in an additional 4.13% improvement in overall mIoU. Finally, we employ threshold filtering and integrate CRF to derive the initial pseudo masks. By refining these masks with insights from the generic SAM, enhancing target boundaries, and rectifying misactivations, the JOS strategy results in the mIoU increase of 5.93%.

Fig. 14 showcases the CAM visualizations at different ablation stages. Initially, the baseline struggles to pinpoint dense small targets (LV) accurately. As we introduce the modules incrementally, the CAM's activated areas become progressively precise. However, the final threshold application across all categories causes irregularities in the generated pseudo masks. As shown in Fig. 15, we utilize the JOS strategy to refine the initial pseudo masks, aligning semantic boundaries with geometric ones for improved accuracy and clarity.

2) Loss Functions: To validate the effectiveness of our loss functions, we sequentially integrate the improved classification loss (cls), contrastive loss (ct), and consistency constraint (cc) into the model after introducing the AAF module.

As depicted in the Table X, it demonstrates an accuracy improvement of the pseudo masks to 32.81% by training our model with the initial multilabel classification loss, signifying a substantial advancement over the baseline. Our GLAM is validated through this improvement. Within our framework, class scores are calculated utilizing both class tokens and patch tokens. To mitigate redundancy, we balance the classification loss by averaging scores from patch tokens and the final optimized CAM_f through average pooling. Addressing the imbalanced data category distribution, we utilize an asymmetric loss to enhance the classification loss L_{cls}, resulting in the mIoU of 35.05%. This loss exclusively influences the output class tokens of the final layer, given that intermediate or shallow layers might lack sufficient guidance for distinct

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TABLE VIII
COMPARISON OF SEGMENTATION RESULTS ON THE VAIHINGEN, MEASURED IN TERMS OF MIOU (%), OA (%), AND F_1 (%)

Method	Imp.suf.	Building	Low Veg.	Tree	Car	mIoU	mF1	OA
Affinitynet	*64.05*	*64.70*	*33.65*	*59.54*	1.01	*44.59*	*56.33*	*91.43*
Irnet	8.95	8.29	3.70	4.59	20.13	9.13	16.23	76.33
SEAM	49.67	47.09	10.51	47.57	7.01	32.37	45.40	87.91
Advcam	3.35	2.43	12.73	5.39	0.42	4.86	8.93	75.82
Mctformer	35.18	54.70	31.14	53.74	8.95	36.75	51.32	88.32
WeakTr	34.73	58.51	30.89	45.93	9.47	35.91	50.57	88.06
Ours	**68.54**	**80.26**	**62.34**	**65.27**	**54.36**	**66.15**	**75.39**	**92.95**

TABLE IX
ABLATION EXPERIMENTS OF MAIN COMPONENTS ON THE iSAID, MEASURED IN TERMS OF MIOU (%)

Method	BG	SH	ST	BD	TC	BC	GTF	BR	LV	SV	HC	SP	RA	SBF	PL	HA	mIoU
baseline	81.59	15.97	15.64	41.58	45.61	26.70	11.33	9.83	10.82	4.83	9.73	22.65	5.95	11.36	40.52	9.56	22.72
+GLAM	82.93	18.72	30.05	46.41	51.21	45.28	24.31	13.13	23.32	11.15	12.36	29.55	12.26	41.96	43.07	20.42	31.63
+AAF	83.41	21.32	33.11	48.57	53.34	52.19	26.38	16.69	26.53	14.22	14.67	30.12	13.59	48.37	52.10	21.67	34.76
+CSC	86.62	31.19	36.21	54.35	61.51	53.87	**39.96**	20.30	32.39	18.01	18.31	36.49	18.27	43.32	45.42	26.03	38.89
+CRF	87.60	32.05	37.37	56.18	61.54	54.30	39.01	21.33	34.38	18.28	18.93	38.56	17.92	42.71	51.25	27.08	39.91
+JOS	**90.10**	**35.43**	**47.10**	**64.37**	**70.10**	**60.17**	38.10	**26.55**	**41.10**	**29.34**	**22.17**	**45.31**	**26.34**	**46.37**	**55.13**	**31.27**	**45.84**

TABLE X
IMPACT OF DIFFERENT LOSSES ON OUR MODEL, MEASURED IN TERMS OF MIOU (%)

L_{src}	L_{cls}	L_{ct}	L_{cc}	mIoU
✓				32.81
	✓			35.05
	✓	✓		37.53
	✓	✓	✓	38.89

TABLE XI
IMPACT OF DIFFERENT LOSSES BASE CLASSIFICATION LOSS, MEASURED IN TERMS OF MIOU (%)

L_{base}	L_c	L_p	L_f	mIoU
✓				22.72
	✓			23.65
	✓	✓		24.01
	✓		✓	23.79
	✓	✓	✓	24.48

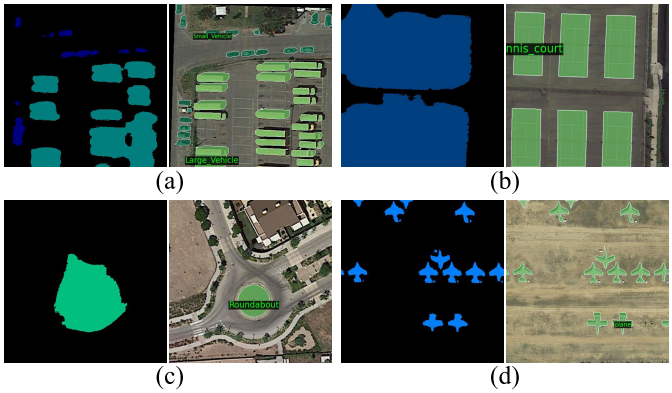


Fig. 15. Pseudo mask obtained after JOS is applied is overlaid onto the original image. In (a), the LV category demonstrates improved boundary clarity and more effective coverage of the target category post-optimization. In (b) and (c), the optimized masks of TC and RA display connected boundaries, addressing the issue of incomplete coverage beyond the target boundaries. In (d), the PL category demonstrates clearer target outlines.

class token differentiation. The introduction of the contrastive loss L_{ct} enhances the mIoU of the entire pseudo masks by 2.48%. Finally, the inclusion of the self-supervised consistency constraint L_{cc} manages to address target scale variations and misactivations culminate in the mIoU of 38.89%.

In addition, we conduct a detailed analysis of the three components within the classification loss. Starting with the baseline L_{base}, we validate the effectiveness of our classification loss, where L_c, L_p, and L_f represent the losses associated with the class token, patch token, and final CAM components, respectively. Considering the inherent class imbalance in RS scenery, Table XI demonstrates that our utilization of the asymmetric loss for improving multilabel losses helps alleviate this imbalance issue. The initial improvement in mIoU is observed. Subsequently, by incorporating multilabel losses separately for the patch token component L_p and the final CAM components L_f, we further improve the metrics. This provides evidence for the effectiveness of the improvements made to the three components within our classification loss, resulting in a 1.76% increase in mIoU over the baseline.

3) Loss Weights: This section examines the impact of different loss weights on mF1, as depicted in Fig. 16. We note that achieving higher mF1 in localization occurs when consistency and category constraints are relatively equivalent. Specifically, setting the weights for L_{cls}, L_{cc}, and L_{ct}, as λ_1, λ_2, and λ_3 to 0.15, 0.425, and 0.425, respectively, results in the highest mF1.

F. Hyperparameter Analysis

1) Background Threshold: We investigate the impact of background thresholding on the initial pseudo mask. Diverse

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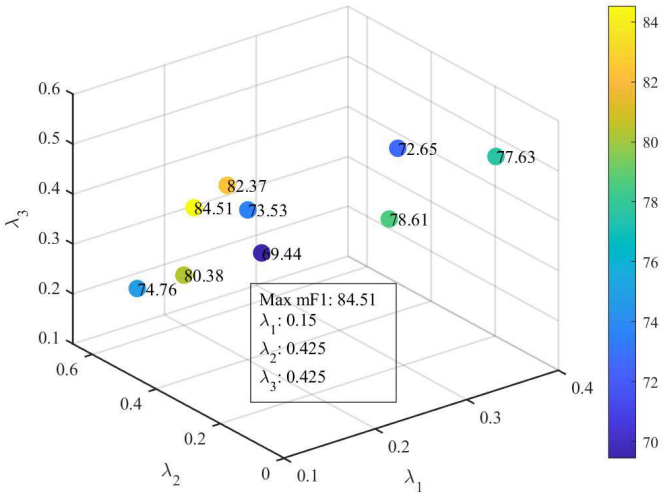


Fig. 16. Influence of weight coefficients on the losses.

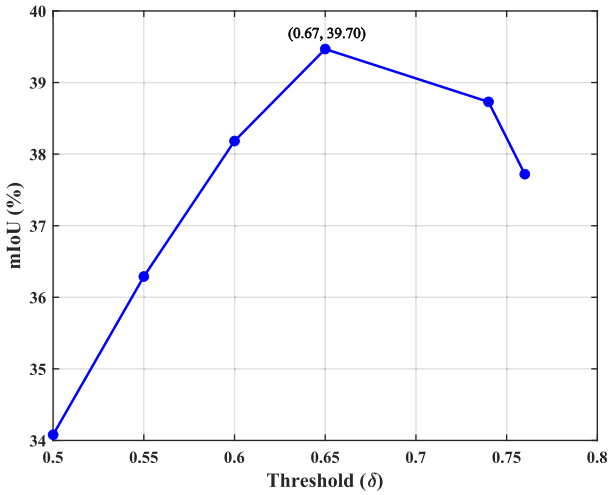


Fig. 17. Impact of background threshold on pseudo mask.

thresholds used in generating the initial pseudo mask yield masks of differing quality. Due to the absence of a background class in the Potsdam dataset, we employ the iSAID dataset for illustration. Fig. 17 demonstrates that our pseudo mask attains the highest quality at $\delta = 0.67$. This finding underscores the criticality of selecting a suitable background threshold, given its substantial influence on the pseudo mask quality.

V. Conclusion

This article proposes CocoaNet, a WSSS model based on consistency-constrained multiclass attention, aiming to address challenges in RS scenes. We employ a global–local adaptive attention mechanism to capture semantic dependencies and class distinctiveness. By optimizing the initial CAMs, we effectively alleviate interclass interference and manage the distribution of densely clustered small objects. The introduction of a consistency constraint corrects activation inaccuracies. Using a Siamese structure, we address substantial scale variations in RS scenes and enhance the discriminativeness of class-specific features with a class contrast loss. Through a JOS, combining prior knowledge from the generic SAM, we manage to obtain the nuanced target boundaries. The experimental results demonstrate the superiority of our method in multiclass RS scenarios compared to SOTA approaches.

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